BRIGHTWATER POLYTECHNIC · ESTATES & CAPITAL PROJECTS

The Halden Engineering & Innovation **Hub**

RIBA Stage 2 concept design and building systems integration proposal for a 7,140 m² net-zero teaching and research building.

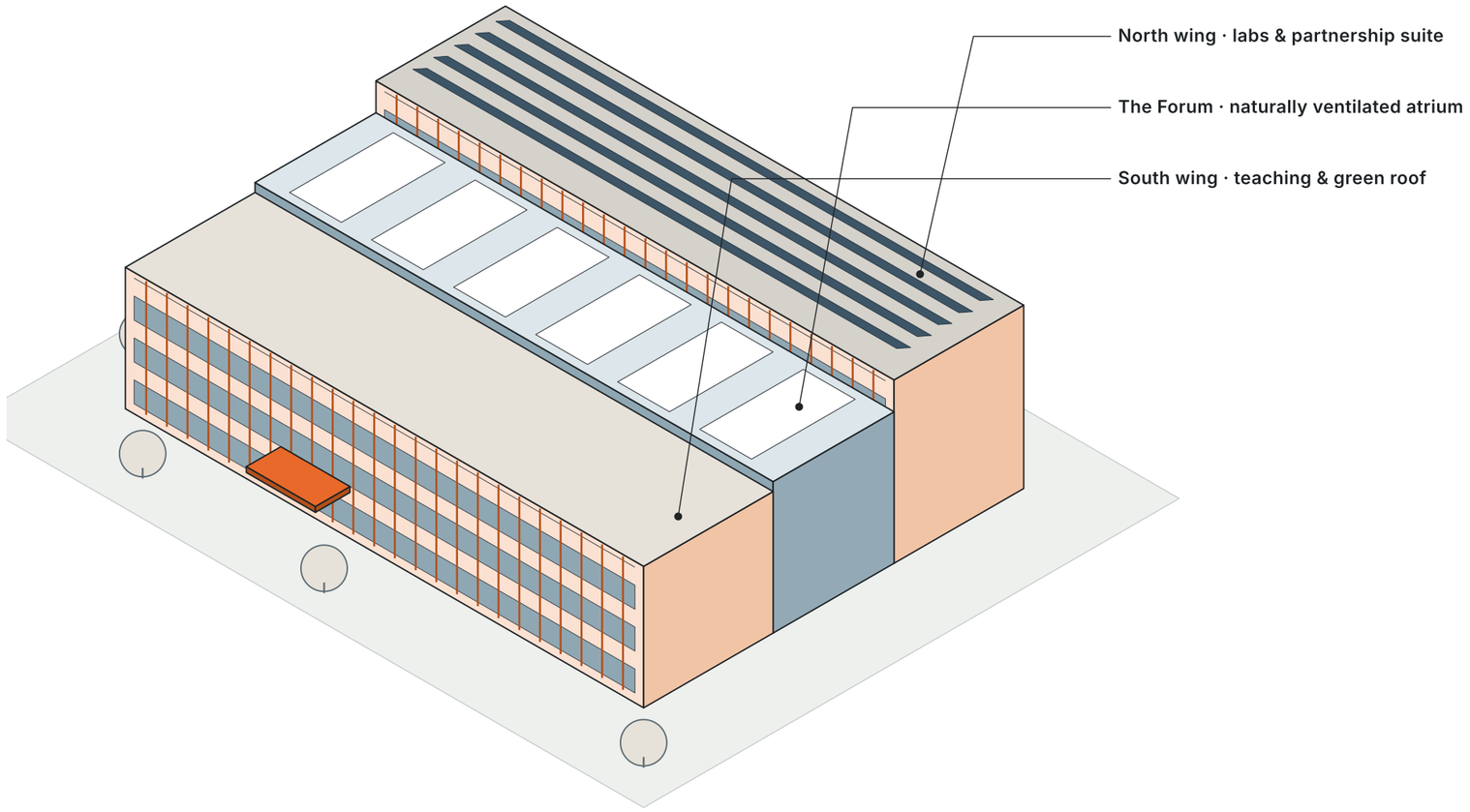


FIG. 01 — AXONOMETRIC FROM THE SOUTH-WEST · CONCEPT MASSING

7,140 m²	£41.8m
Gross internal area	Construction budget (Q3 2026 prices)
62	Sept 2029
kWh/m²·yr operational energy target	First teaching term

COMPLETED 4 SEPTEMBER 2026

Designed by Cynthia Agwaraonye

PROJECT	CLIENT	SHEET	SCALE @ A3	STAGE	DATE	NO.
Halden Engineering & Innovation Hub	Brightwater Polytechnic	Cover	NTS	RIBA 2 · Rev C	04.09.2026	HEIH-000

One building for teaching, **making** and industry partnership

Three 1960s buildings become one hub for **1,900 students**, 140 staff and an industry partnership suite, on a brownfield car-park site.

Client objectives

#	OBJECTIVE	DESIGN RESPONSE
01	Make engineering visible and social	Glazed Forum atrium links every lab to the campus street
02	Net zero carbon in operation	All-electric heat pumps, 310 kWp PV, Passivhaus-informed fabric
03	Flexible for 40 years of curriculum change	7.5 m structural grid, raised floors, plug-and-play lab services
04	Showcase smart building technology	Open-protocol BMS and digital twin used as a teaching resource
05	Deliver within £41.8m and by summer 2029	Hybrid CLT–steel frame with offsite-manufactured cores

1,900
Students served

4
Storeys (north wing), 3 south

BREEAM
Outstanding (target, 2018 NC)

−58%
Embodied carbon versus RIBA 2025 business-as-usual

2.1 ha
Site area, brownfield

+15%
Biodiversity net gain

DESIGN PRINCIPLES

P1 · Show the work
Labs face the Forum through glazed screens, so the teaching is visible to anyone passing through.

P2 · Fabric first
Compact form, high airtightness and fixed shading cut demand before any technology is added.

P3 · Loose fit
Long-span floors and services zones let rooms change use without structural change.

P4 · Honest materials
Exposed timber, recycled brick and terrazzo are durable, low-carbon and tactile.

ACCOMMODATION MIX · 7,140 M² GIA



ENGAGEMENT TO DATE

38
stakeholder workshops

412
student survey responses

6
industry partners consulted

2
planning pre-application meetings

PROJECT	CLIENT	SHEET	SCALE @ A3	STAGE	DATE	NO.
Halden Engineering & Innovation Hub	Brightwater Polytechnic	Brief & vision	NTS	RIBA 2 · Rev C	04.09.2026	HEIH-001

A riverside site between **campus** and town

2.1 ha brownfield site between the River Halden and Campus Street.

Opportunities

+ Long east–west axis, ideal south light
+ 11 mature trees retained
+ ~6,000 people a day pass the entrance
+ River frontage for a public terrace

Constraints

– 30 m flood-zone set-back
– Made ground needs remediation
– HGV access kept off pedestrian routes
– 20 m masterplan height cap

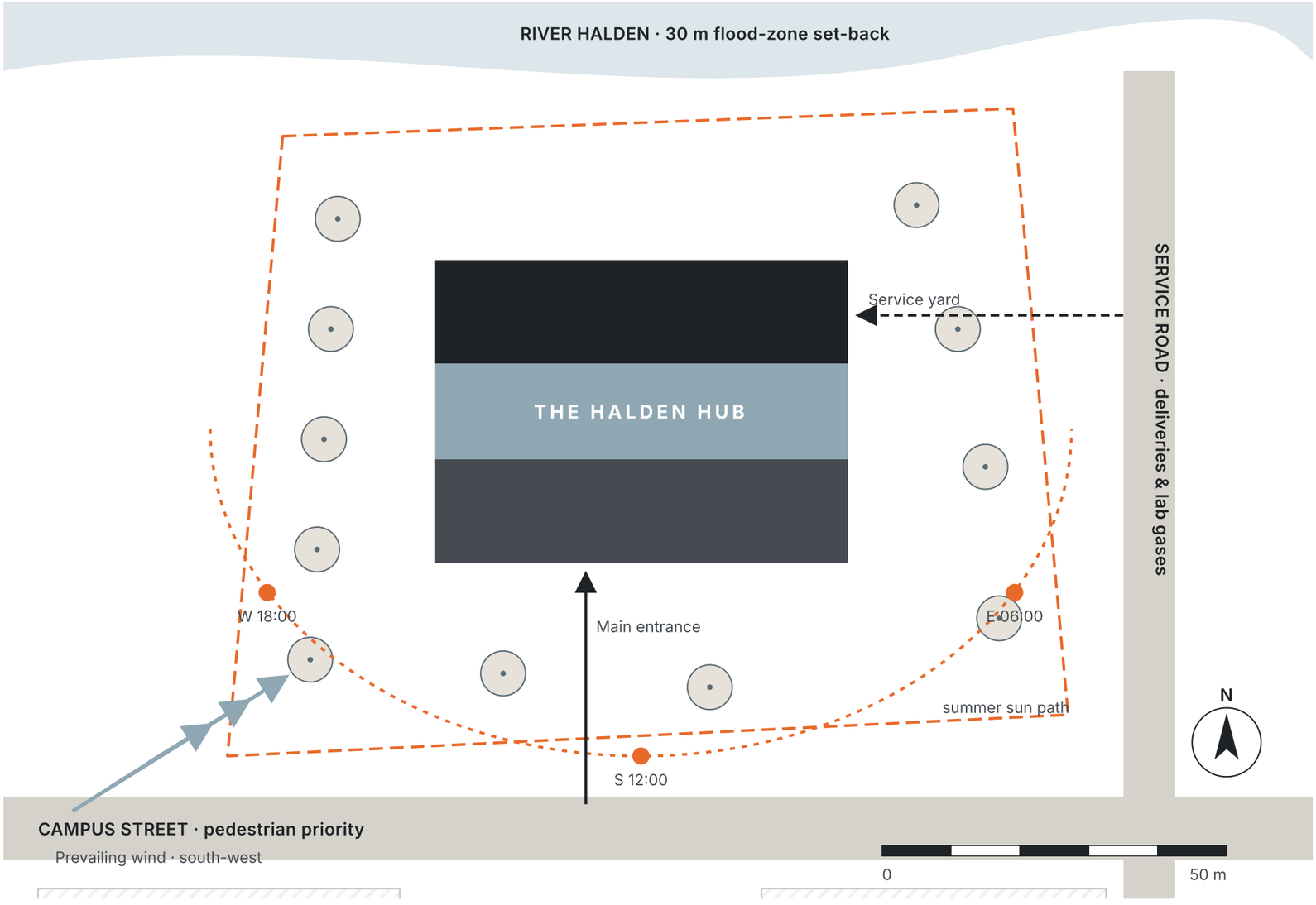
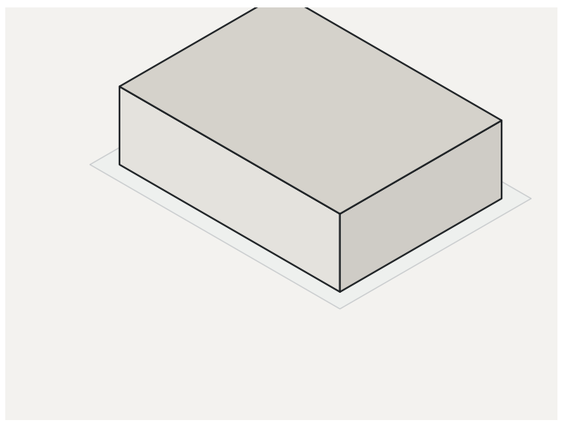


FIG. 02 — SITE PLAN · ORIENTATION, ACCESS, SUN PATH AND PREVAILING WIND

PROJECT	CLIENT	SHEET	SCALE @ A3	STAGE	DATE	NO.
Halden Engineering & Innovation Hub	Brightwater Polytechnic	Site analysis	1:1000	RIBA 2 · Rev C	04.09.2026	HEIH-002

From site envelope to a building **shaped by daylight**

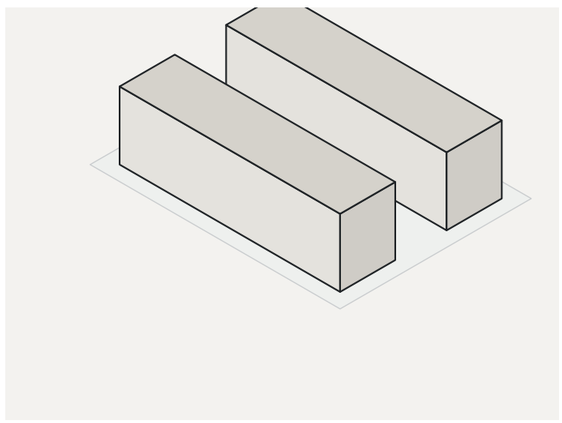
Daylight, airflow and encounter shape the massing. No step adds area.



01

Site envelope

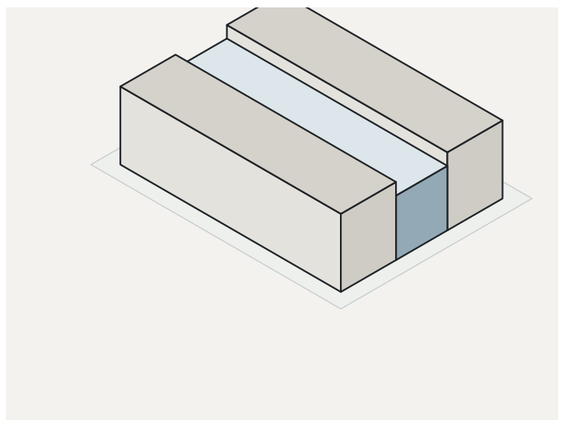
60 × 44 × 18 m: too deep for daylight.



02

Split into two wings

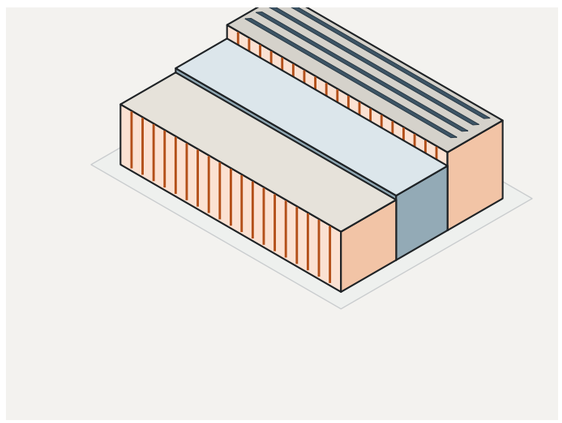
15 m-deep wings: daylight and cross-ventilation.



03

Carve the Forum

14 m glazed atrium: social and ventilation heart.



04

Tune to climate

South wing drops a storey; PV, green roof, timber fins.

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15 m

Wing depth: daylight factor ≥ 3% to 92% of floor area

14 m

Forum width for stack-effect ventilation

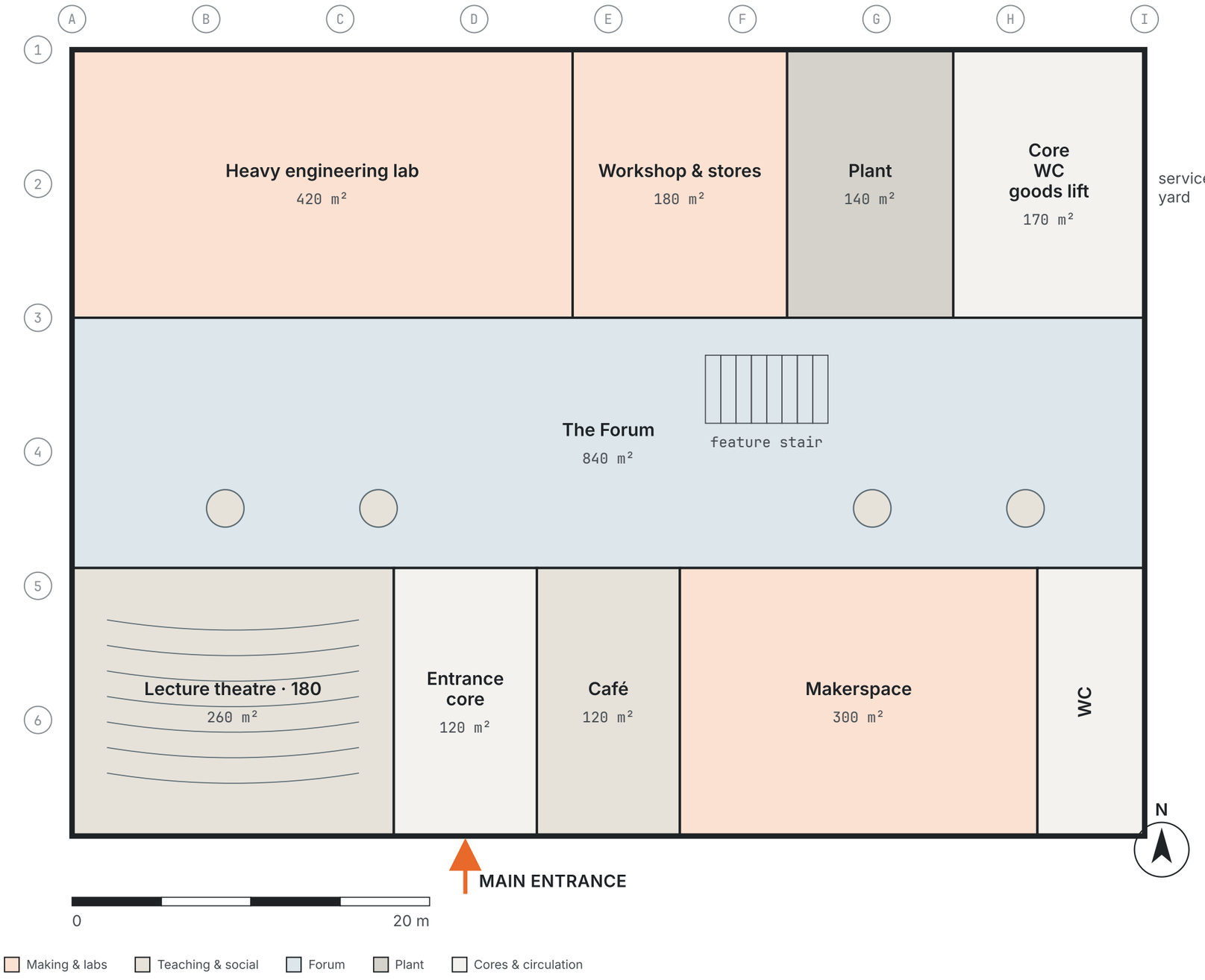
40%

Window-to-wall ratio on south and north façades

0.29

Form factor (heat-loss area to floor area), compact for its type

PROJECT	CLIENT	SHEET	SCALE @ A3	STAGE	DATE	NO.
Halden Engineering & Innovation Hub	Brightwater Polytechnic	Concept massing	NTS	RIBA 2 · Rev C	04.09.2026	HEIH-010



04 · GROUND FLOOR PLAN

Heavy and noisy at the back, **public** at the front

— **North wing:** heavy lab and workshops beside the service yard, on view from the Forum
— **South wing:** lecture theatre, café and makerspace open onto Campus Street

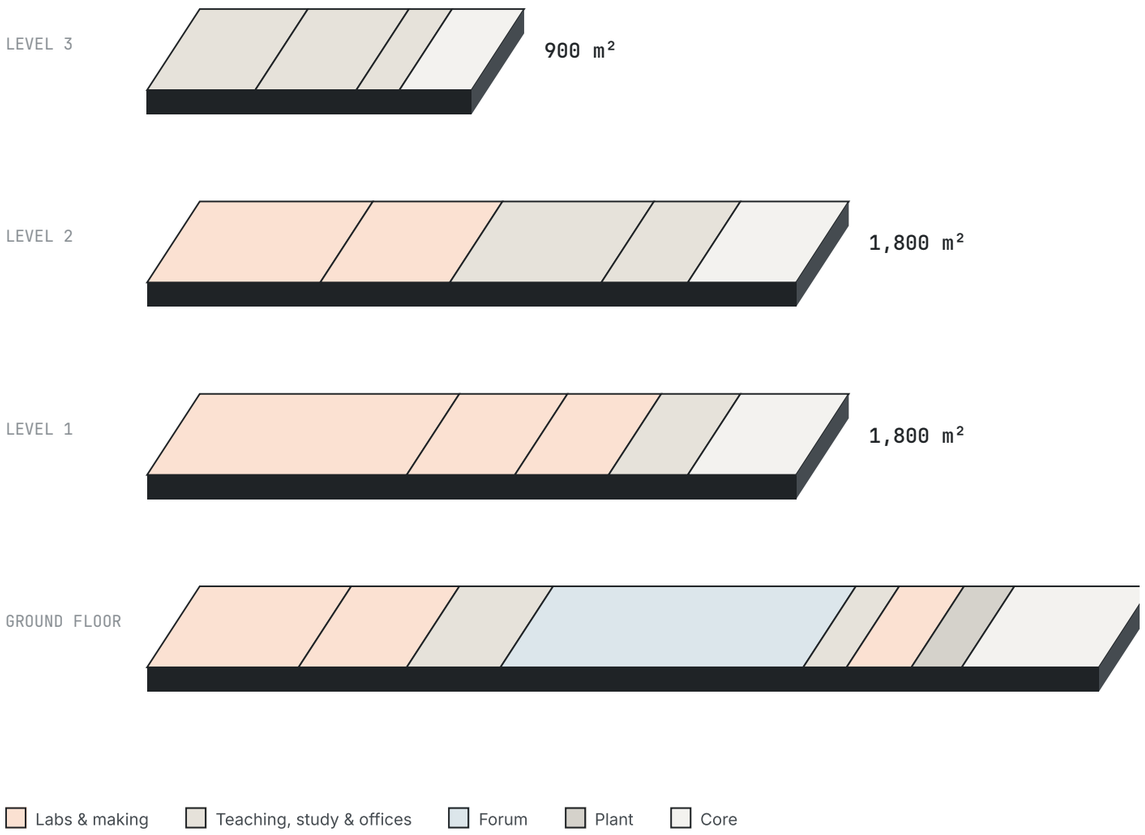
Planning grid

7.5 m grid (A–I, 1–6) suits lab benching and efficient CLT spans.

GROUND FLOOR	M²
Heavy engineering lab	420
Makerspace	300
Lecture theatre (180)	260
The Forum	840
Workshop & stores · Café · Plant	440
Cores & circulation	380
Total ground floor GIA	**2,640**

PROJECT	CLIENT	SHEET	SCALE @ A3	STAGE	DATE	NO.
Halden Engineering & Innovation Hub	Brightwater Polytechnic	Ground floor plan	1:500	RIBA 2 · Rev C	04.09.2026	HEIH-100

Noise and load lowest down, **quiet and partnership** work at the top



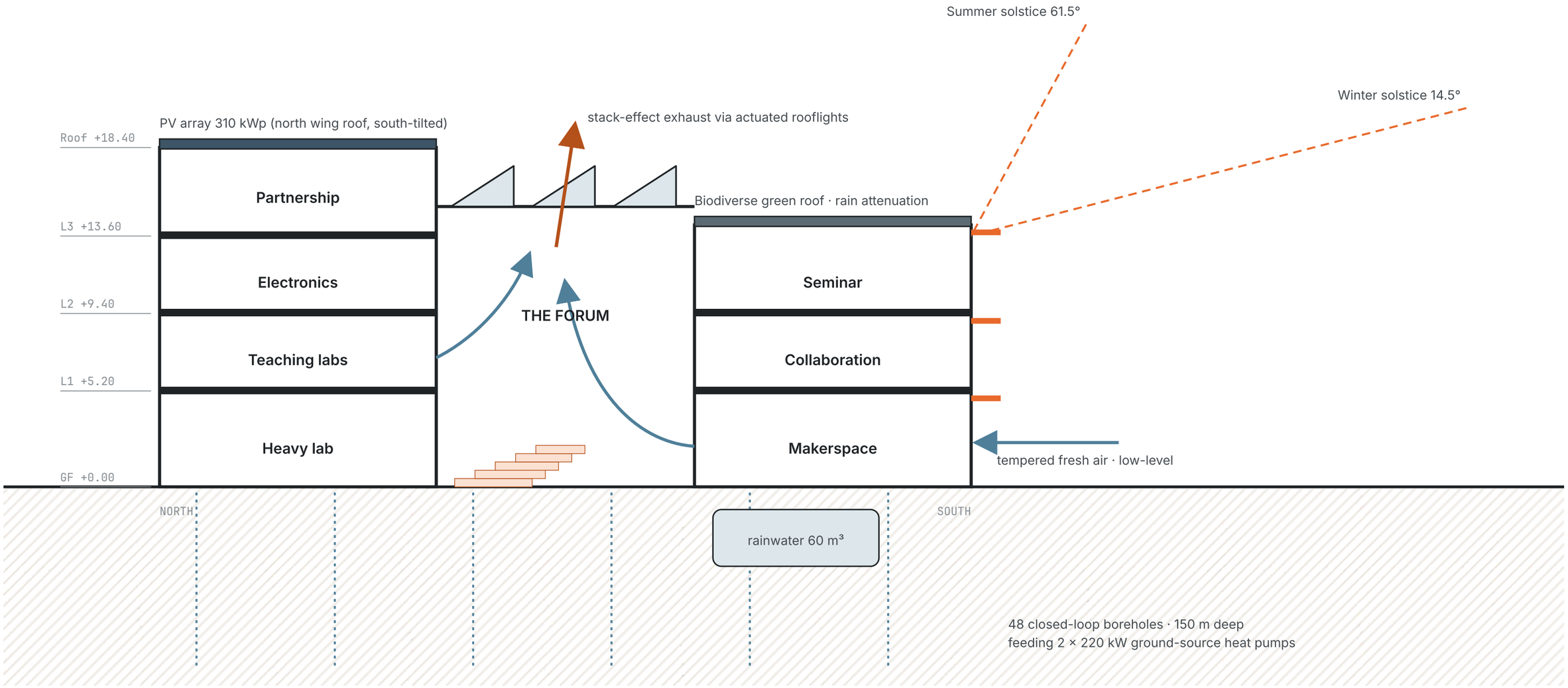
LEVEL	SPACE	GIA M²
Level 3	Industry partnership suite	300
Level 3	Research offices	280
Level 3	Roof terrace & meeting	120
Level 3	Core & circulation	200
	Subtotal	**900**
Level 2	Electronics labs	480
Level 2	Computer suites	360
Level 2	Seminar rooms (6)	420
Level 2	Study space	240
Level 2	Core & circulation	300
	Subtotal	**1,800**
Level 1	Teaching labs (4)	720
Level 1	Robotics & autonomy lab	300
Level 1	Collaboration space	260
Level 1	Staff offices	220
Level 1	Core & circulation	300
	Subtotal	**1,800**
Ground floor	Heavy engineering lab	420
Ground floor	Makerspace	300
Ground floor	Lecture theatre	260
Ground floor	The Forum	840
Ground floor	Café	120
Ground floor	Workshop & stores	180
Ground floor	Plant	140
Ground floor	Core & circulation	380
	Subtotal	**2,640**
	Total gross internal area	**7,140**

Net-to-gross efficiency 74% excluding the Forum. Areas measured to RICS IPMS 1 and rounded to 10 m².

PROJECT	CLIENT	SHEET	SCALE @ A3	STAGE	DATE	NO.
Halden Engineering & Innovation Hub	Brightwater Polytechnic	Stacking & areas	NTS	RIBA 2 · Rev C	04.09.2026	HEIH-110

The Forum **breathes** for the whole building

Eight months a year the Forum is a solar chimney: air in low on the south façade, out through actuated rooflights.



8 months
Natural ventilation season (mixed-mode)

440 kW
Ground-source heating and cooling capacity

0.6
Air permeability target (m³/h·m² at 50 Pa)

67%
Annual electricity demand matched by on-site PV

40%
Potable water saving from rainwater reuse

PROJECT	CLIENT	SHEET	SCALE @ A3	STAGE	DATE	NO.
Halden Engineering & Innovation Hub	Brightwater Polytechnic	Section A–A · north-south	1:250	RIBA 2 · Rev C	04.09.2026	HEIH-200

Open protocols, one data model, **one twin**

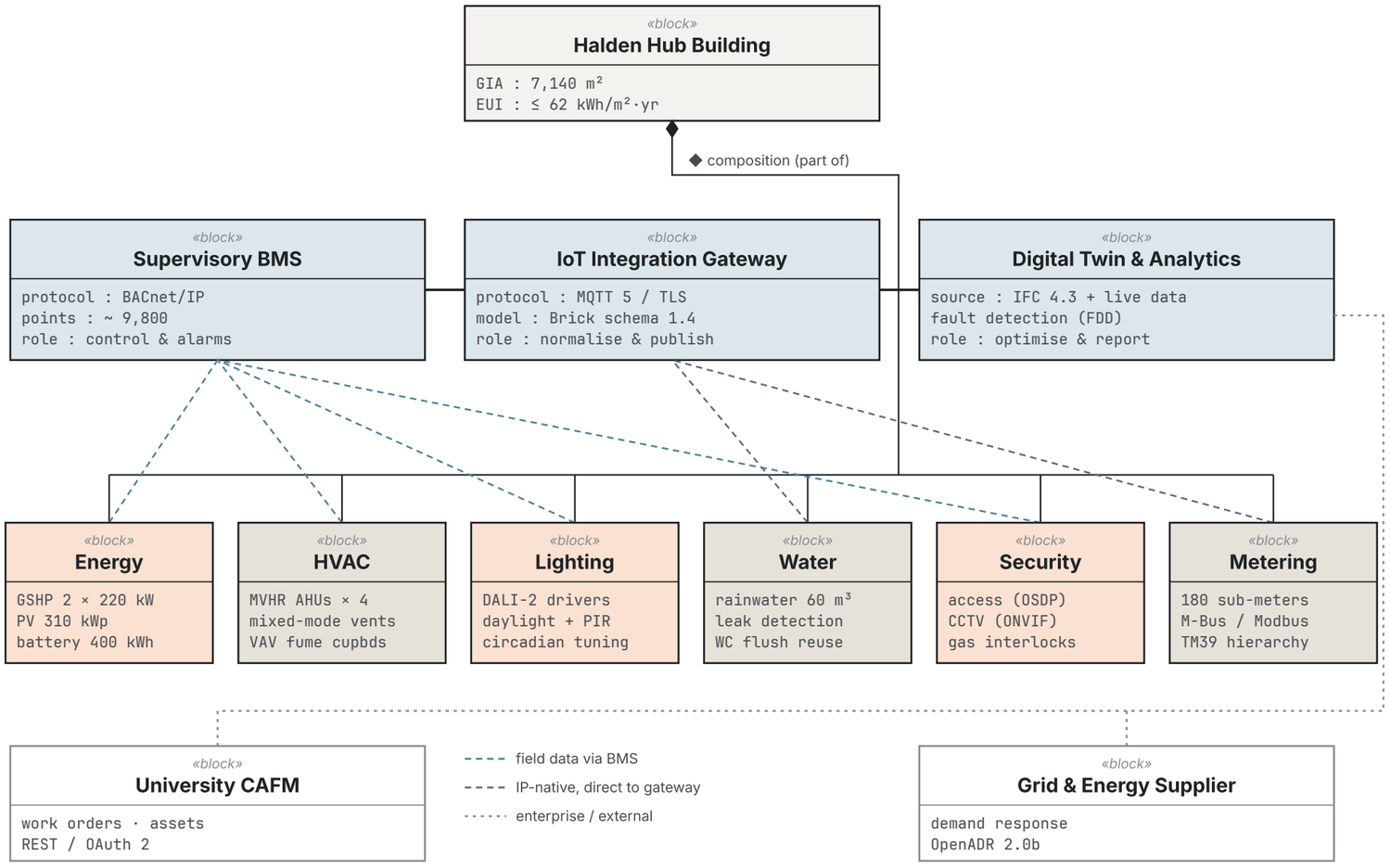


FIG. 07 — SYSML BLOCK DEFINITION DIAGRAM · BUILDING SYSTEM ARCHITECTURE (CONCEPT LEVEL)

Every subsystem speaks open protocols into one data model: run, optimised and taught from the same live data.

Interface control summary

ID	FROM → TO	PROTOCOL · DATA
IF-01	Energy subsystem → Supervisory BMS	**BACnet/IP** Heat-pump state, COP, PV yield, battery state of charge
IF-02	Lighting subsystem → Supervisory BMS	**DALI-2 → BACnet** Occupancy, lux levels, driver faults, energy per zone
IF-03	Supervisory BMS → IoT gateway	**MQTT 5 / TLS 1.3** All points tagged with Brick 1.4 semantics
IF-04	Metering subsystem → IoT gateway	**Modbus TCP → MQTT** Half-hourly sub-meter readings (TM39)
IF-05	Digital twin → University CAFM	**REST · OAuth 2** Fault-driven work orders, asset condition
IF-06	Digital twin → Energy supplier	**OpenADR 2.0b** Demand-response events, load-shift schedules

Requirements traced to Stage 3

42 requirements (function, performance, IEC 62443-3-3 SL2 security), each traced to a block and an interface.

PROJECT	CLIENT	SHEET	SCALE @ A3	STAGE	DATE	NO.
Halden Engineering & Innovation Hub	Brightwater Polytechnic	Systems integration	NTS	RIBA 2 · Rev C	04.09.2026	HEIH-300

All-electric, **ground-coupled**, grid-flexible

One electrical spine feeds heat pumps on a shared ground loop. PV and the battery shift load away from grid peaks.

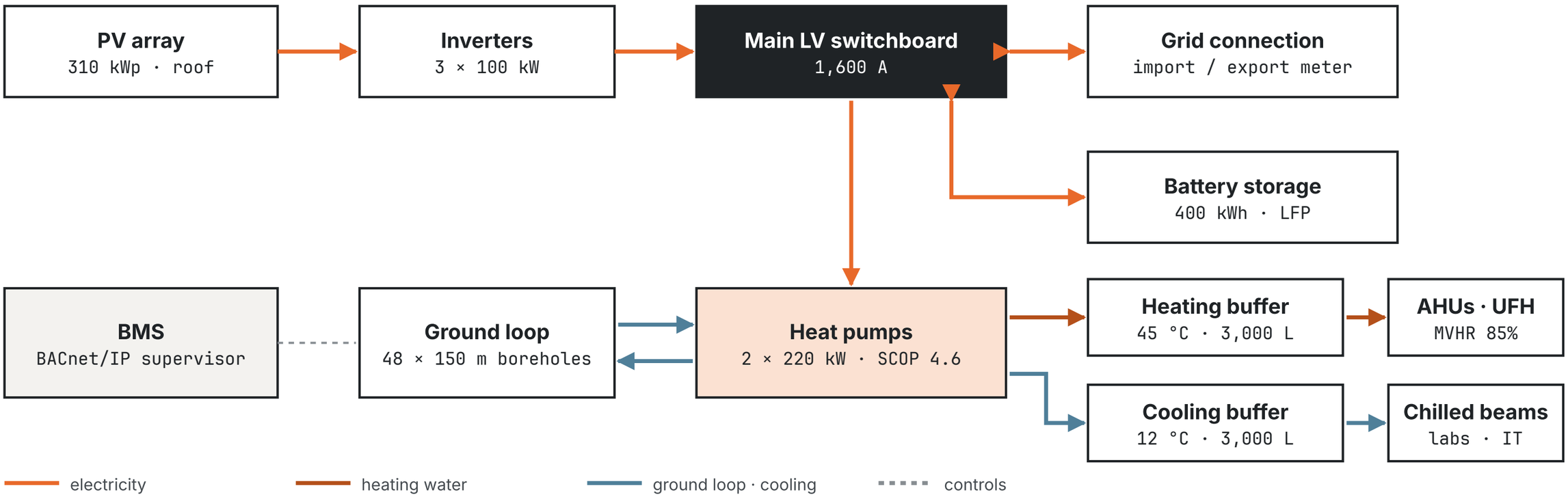


FIG. 07B — ENERGY PLANT SCHEMATIC · CONCEPT LEVEL

0	4.6	295 MWh	−38%	85%
fossil fuel on site	heat-pump seasonal COP	annual PV yield	peak grid demand with battery	MVHR heat recovery

Plant schedule

TAG	ITEM	CAPACITY
GSHP-01/02	Ground-source heat pumps	2 × 220 kW heat · 2 × 180 kW cool
GL-01	Closed ground loop	48 boreholes × 150 m
PV-01	Photovoltaic array	310 kWp · 295 MWh/yr

TAG	ITEM	CAPACITY
BAT-01	Battery energy storage	400 kWh LFP · 200 kW
AHU-01–04	Air handling units (MVHR)	4 × 6 m³/s · 85% recovery
BV-01/02	Buffer vessels	2 × 3,000 L · 45 °C / 12 °C

PROJECT	CLIENT	SHEET	SCALE @ A3	STAGE	DATE	NO.
Halden Engineering & Innovation Hub	Brightwater Polytechnic	Energy plant schematic	NTS	RIBA 2 · Rev C	04.09.2026	HEIH-310

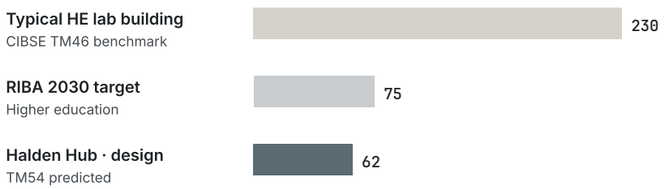
Net zero in operation, **low carbon** to build

UK Net Zero Carbon Buildings Standard (pilot) and RIBA 2030 targets, verified by TM54 modelling.

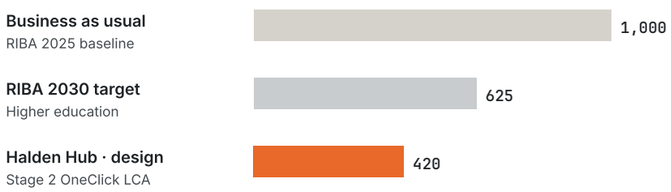
Key strategies

01 Passivhaus-informed fabric · U 0.12 walls, 0.10 roof
02 All-electric: ground-source heat pumps
03 CLT + steel frame, 50% GGBS concrete
04 Reclaimed brick plinth
05 400 kWh battery + demand response

OPERATIONAL ENERGY · KWH/M²·YR



EMBODIED CARBON (A1–A5) · KGCO₂E/M²



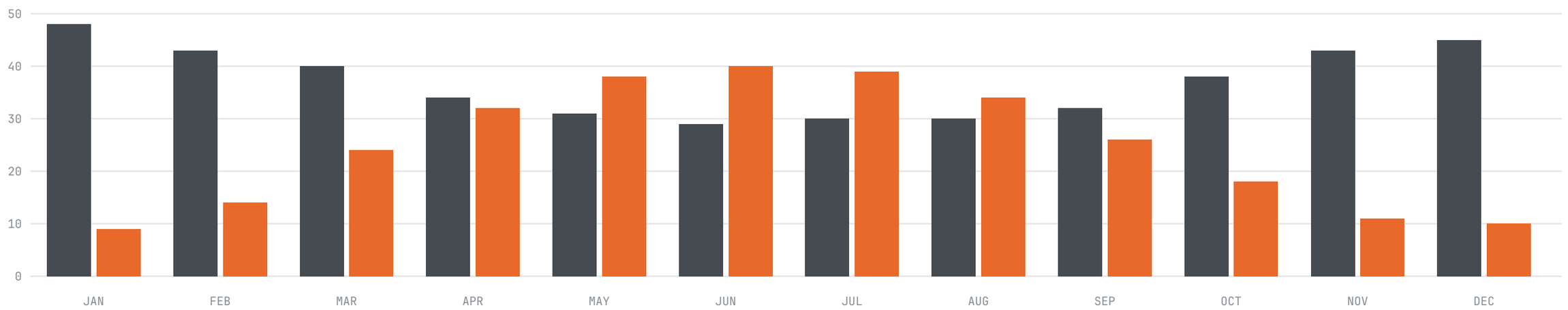
CERTIFICATION & TARGETS

METRIC	TARGET
BREEAM New Construction 2018	Outstanding
Operational energy (EUI)	62 kWh/m²
Space-heating demand	15 kWh/m²
Upfront embodied carbon	420 $kgCO_2e/m^2$
On-site renewable generation	295 MWh/yr
Potable water use	−40%
Biodiversity net gain	+15%
WELL v2 (Forum & teaching)	Gold-ready

POST-OCCUPANCY COMMITMENT

Three years of Soft Landings aftercare (BSRIA BG 54). Metered performance will be published annually through the digital twin dashboard.

MONTHLY ENERGY BALANCE · MWH · TM54 PREDICTION



PROJECT	CLIENT	SHEET	SCALE @ A3	STAGE	DATE	NO.
Halden Engineering & Innovation Hub	Brightwater Polytechnic	Sustainability	NTS	RIBA 2 · Rev C	04.09.2026	HEIH-400

09 · MATERIALS

A palette that is **warm, robust** and low-carbon

Chosen to take hard use, be repaired, and show how the building is made.

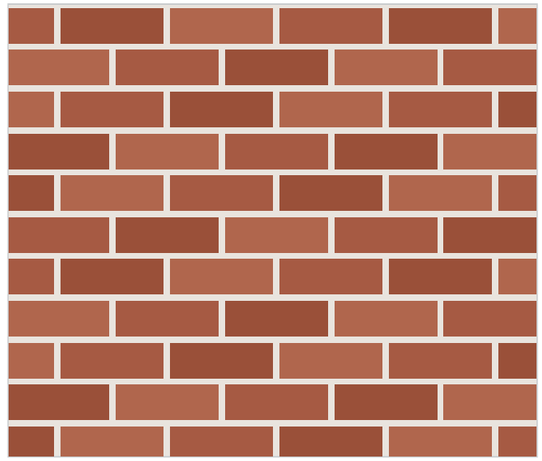
— Timber exposed wherever fire strategy allows
— Hard, recycled finishes where wear is highest
— PEFC/FSC timber, tracked in a material passport



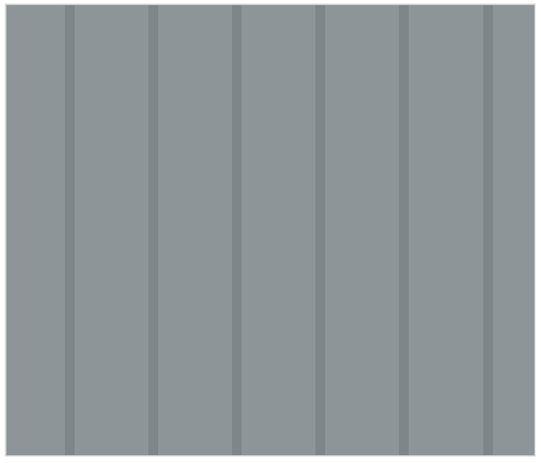
M01
Cross-laminated timber
Exposed soffits and walls, spruce, PEFC-certified



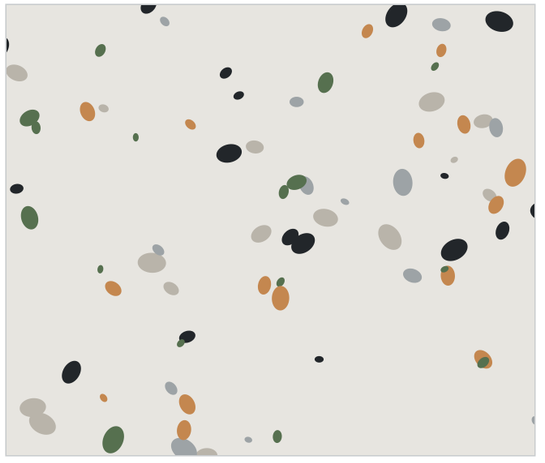
M02
Glulam frame
Forum columns and roof beams, larch



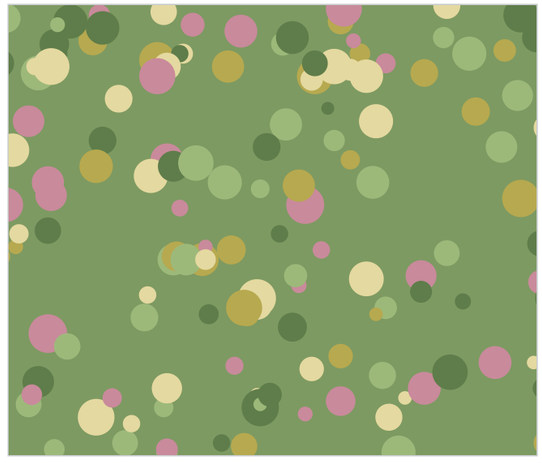
M03
Reclaimed brick
Ground-floor plinth, reclaimed from site



M04
Anodised aluminium
Rainscreen & window frames, 75% recycled



M05
Recycled-aggregate terrazzo
Forum floor, including crushed site concrete



M06
Biodiverse roof
South wing roof, native wildflower mix

PROJECT	CLIENT	SHEET	SCALE @ A3	STAGE	DATE	NO.
Halden Engineering & Innovation Hub	Brightwater Polytechnic	Materials palette	NTS	RIBA 2 · Rev C	04.09.2026	HEIH-500

From concept to the **first teaching term** in three years

Mar–Sep 2026
Stage 2 · Concept
This report. Concept approved by the Capital Programme Board.

Oct 2026–Feb 2027
Stage 3 · Spatial coordination
Systems requirements baselined, BIM to LOD 300.

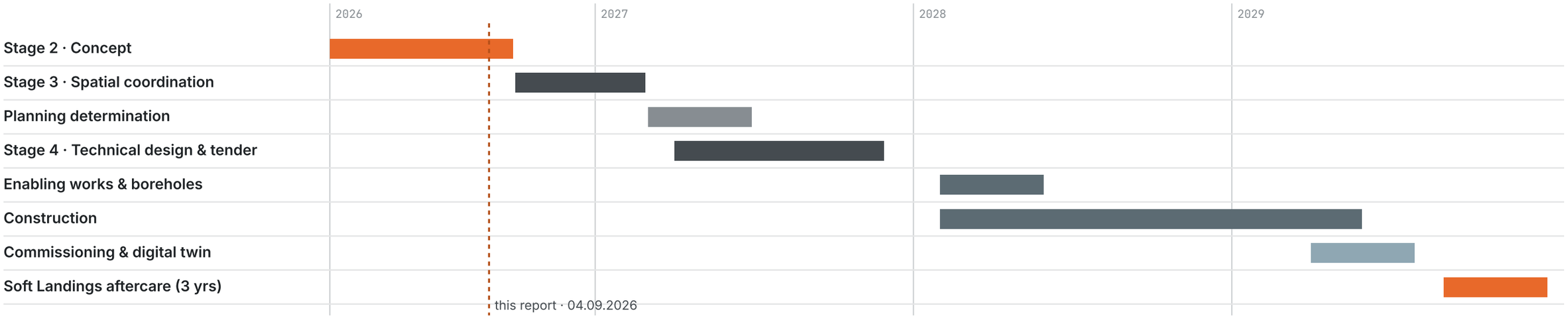
Mar 2027
Planning submission
Pre-application meetings held May and July 2026.

Apr–Nov 2027
Stage 4 · Technical design
Two-stage design and build tender.

Feb 2028
Start on site
Remediation and borehole drilling first.

Jun 2029
Practical completion
Seasonal commissioning; digital twin goes live.

Sep 2029
First teaching term
Soft Landings aftercare begins.



Decisions requested
1. Approve the Stage 2 concept and the £41.8m cost plan.
2. Confirm the open-protocol BMS and digital-twin strategy, procured separately from the main contract.
3. Release £2.6m of Stage 3 fees.

Key risks
Ground contamination worse than surveyed (contingency held). CLT supply-chain lead times (early contractor involvement). Planning conditions on river-edge biodiversity.

DOCUMENT CREDITS

Brightwater Polytechnic · Estates & Capital Projects
Halden Engineering & Innovation Hub
RIBA Stage 2 Concept Design Report · Rev C
Completed 4 September 2026

Designed by Cynthia Agwaraonye

Fictional client and illustrative data, prepared as a portfolio case study. Drawings are concept-level and not for construction.